Wazuh Implementation

Isaac Reyes

Computer networks face constant attacks. They go unnoticed until it is too late. How do you defend against these attacks?

Project Goal

Implement a Wazuh server and agent that will detect a brute force attack and respond to the attack.

Wazuh

Wazuh is an open-source Security Incident Event Management (SIEM) program that includes log collection, rule-based detection, and active responses.

Architecture

The project consists of 3 parts:

- Wazuh server – Sever, indexer, and dashboard
- Wazuh agent – Endpoint that will be attacked
- Kali machine – Simulate a brute force attack

The Attack

An application called Hydra, will be used to attempt to establish a ssh connection utilizing a text file of passwords.

Detection

Endpoints logs are forwarded to the Wazuh server where they are analyzed. When suspicious activity is detected, the dashboard displays the alert in real time, and the data is indexed for review.

Active Response

When an alert is detected, a response script is executed. A detected brute force attack will run a firewall drop script, blocking the attacking IP address for a certain time.

Lessons Learned

I learned the importance of a well-maintained system that provides security. A brute force attack is not something that most machines natively detect. A system that can detect then respond will make for a difficult target to attack.

Conclusion

Computer security is a dynamic field. A strong network of computers may be utilized for good but may also be exploited. A SIEM system can detect attacks and respond rapidly to minimize potential damage.